# 국제 금융 시장 지수의 그래프 신경망 접근법 기반을 통한 수익률 예측을 위한 데이터 효율성 제고

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## 1 Abstract


- This study examines the applicability of graph neural networks, specifically Graph Attention Networks (GAT), for finan cial return prediction across 15 major global stock indices.
- By incorporating inter-market relationships, these models aim to capture spillover effects and nonlinear dependencie s that traditional machine learning models may overlook.
- Through 30 repeated time- series cross-validation experiments, GAT generally achieve lower RMSE and MAE compar ed to benchmark models such as XGBoost, MLP, and SVM, particularly during financial crises, post-crisis recoveries, and volatile periods.
- However, their advantage is less pronounced in stable markets. Model performance was evaluated across 30 experim ents, with statistical significance tested at 1%, 5%, and 10% levels. The findings highlight the potential of graph-based models in financial forecasting while underscoring the need for further research on interpretability and computational efficiency.

## 2 Introduction

- This study proposes a financial forecasting method using Graph Neural Networks (GNNs). GNNs can capture nonline ar dependencies between global financial markets by learning from relationships between nodes, making them well-s uited for modeling inter-market interactions.
- While previous studies have applied GNNs to stock markets, cryptocurrencies, or single-market indices, research on modeling global financial market interactions remains limited.
- Also, while many studies claim improved predictive accuracy, their validation processes are often insufficient, lacking rigorous statistical significance testing.
- To address these limitations, this study applies Graph Attention Networks (GATs) to predict the daily returns of 15 ma jor global financial indices.
- Unlike traditional approaches, our method does not rely on external macroeconomic indicators or sentiment-based d ata. Instead, it leverages intrinsic market data, enhancing data efficiency and minimizing potential biases.



## 3 Literature Review

- Xiang et al. (2023) demonstrated that **GNNs effectively capture temporal dependencies** in stock market prediction, o utperforming traditional ML and DL models.
- Chen et al. (2023) integrated **natural language processing (NLP) with GNNs** to incorporate sentiment analysis into fin ancial forecasting.
- Cheng et al. (2022), Choi & Kim (2024), and Das et al. (2024) attempted to apply **graph-based models to financial risk prediction**.

This study aims to predict financial returns by measuring the spillover effects among global indices by examining wheth er utilizing graph-based embeddings improves predictive performance compared to benchmark models that rely solely o n raw data.

**1. Graph Attention Networks (GAT)**

The **Graph Attention Network (GAT)** enhances Graph Neural Networks by incorporating an **attention mechanism**, allowin g the model to assign different weights to neighboring nodes.

- Calculating the Attention Coefficient

$$e_{ij} = \text{LeakyReLU}\left(a^T\left(Wh_i \,||\, Wh_j\right)\right)$$

where $a^T$ is the attention vector and $W$ is the trainable weight matrix.





- Normalize Attention Coefficient

$$a_{ij} = \text{softmax}(e_{ij})$$

- Aggregate The Features of Neighbor Nodes

$$h_i' = \sigma\left(\sum_{j \in N(i)} \alpha_{ij} W h_j\right)$$

The heads are aggregated by using the multi-head attention method.

**2. Benchmark Models**

**Five benchmark models** are used as a comparison to the graph-based model (GAT).

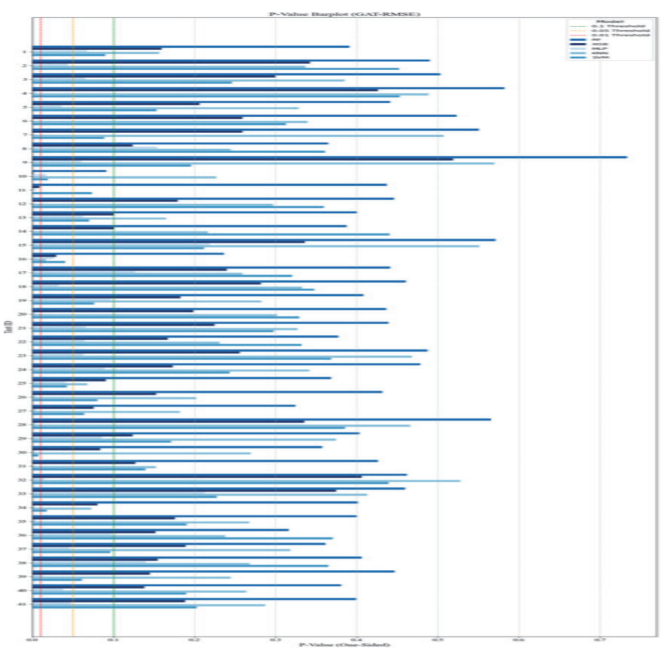
- Random Forest Regressor (RF)
- Gradient Boost Regressor (XGB)
- Multi-Layer Perceptron Regressor (MLP)
- K-Nearest Neighbors Regressor (KNN)
- Support Vector Regressor (SVM)

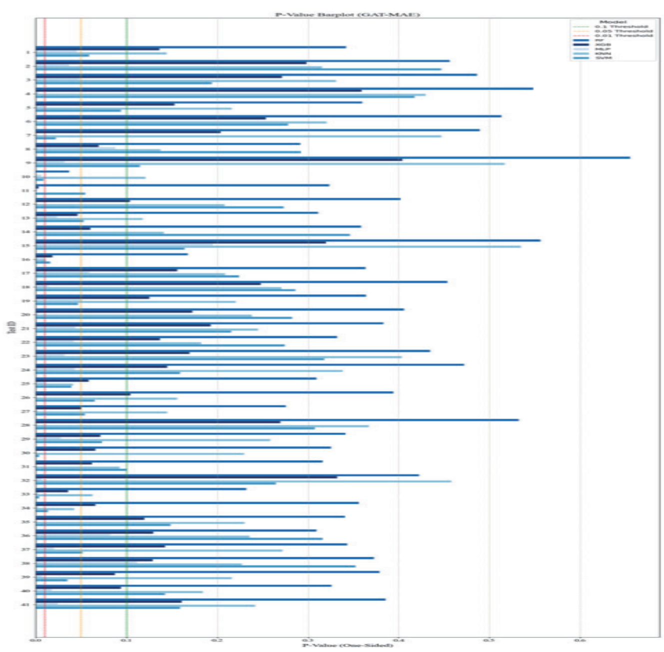
## 5 Dataset

| | |
|---|---|
| Close Price Data | • 2004.01.01 ~ 2024.12.31 , crawling from “Investing.com”<br>• Used for graph data generation |
| Volume Data | • 2004.01.01 ~ 2024.12.31 , crawling from “Investing.com”<br>• Used for selecting 15 indices on market capitalization |
| Graph Data | • Total of 41 test sets, each with 6 months input and 6 months ou tput<br>• The nodes of the graph are the global indices, and the edges ar e created based on the correlation between two indices |



## 6 Results & Conclusion



## 6 Results & Conclusion

- In Test 10 (period 2008 ~ 2009), following the collapse of Lehman Brothers, GAT outperformed XGB(p = 0.0011) and MLP (p = 0.0169), highlighting its ability to capture systemic market shocks.

- In Test 16 (period 2011 ~ 2012), covering heightened uncertainty surrounding Greece and EU bailout negotiations, GAT again outperformed XGB (p = 0.0298), MLP (p = 0.0265), and KNN (p = 0.0172)

- In Test 31 (period 2019 ~ 2020, covering the sharp downturn in the early COVID-19 pandemics), demonstrated GAT's exceptional capability, outperforming MLP in RMSE (p < 0.0001). In Test 33 (period 2021), GAT further distinguished itself by outperforming MLP (p = 0.0029) and SVM (p = 0.0034) in MAE.

In conclusion, the superiority of the graph models mostly stands out during the financial crisis periods.



## 6 References


- Xiang, S., Cheng, D., Shang, C., Zhang, Y., & Liang, Y. (2023). Temporal and Heterogeneous Graph Neural Network for Financial Time Series Prediction. arXiv preprint arXiv:2305.08740.
- Yin, W., Chen, Z., Luo, X., & Kirkulak-Uludag, B. (2024). Forecasting cryptocurrencies' price with the financial stress index: a graph neural network prediction strategy. Applied Economics Letters, 31(7), 630-639.
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**Thank you.**